



CS 540 Introduction to Artificial Intelligence **Logic**

University of Wisconsin-Madison

Spring 2025

Announcements

- **HW 1 online:**
 - Writing assignment---nothing too stressful
 - Deadline **today at 11:59PM**
- **HW 2:**
 - Probability
 - Deadline **Thursday Feb. 6th 11:59PM**

- Class roadmap:

Logic
NLP
Machine Learning: Introduction
Machine Learning: Unsupervised Learning I

} Mostly
Foundations

Logic & AI

Why are we studying logic?

- **Traditional** approach to AI ('50s-'80s)

- ★ – “Symbolic AI” : Explicit, human-readable rules

- The Logic Theorist – 1956

- Proved a bunch of theorems!

- Logic also the language of:

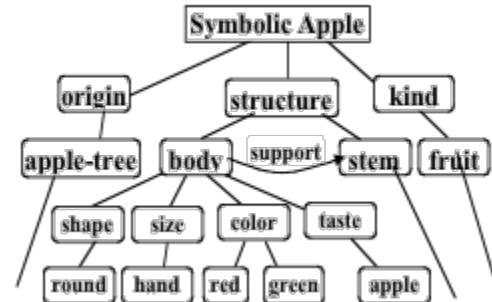
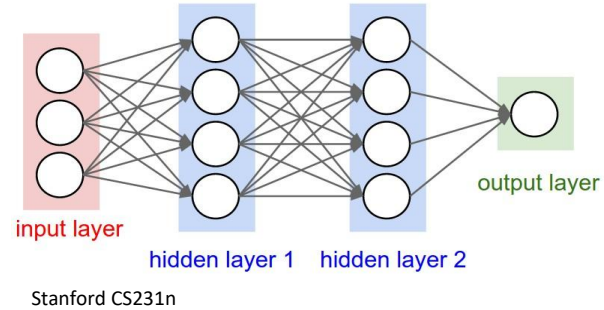
- Knowledge rep., databases, etc.



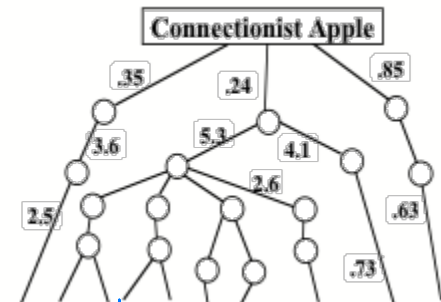
Symbolic vs Connectionist

Rival approach: connectionist

- Probabilistic models ↓
- Neural networks ★ (Focuses on probability and neural networks)
- **Extremely popular** last 20 years



M. Minsky ↓ more logic involved

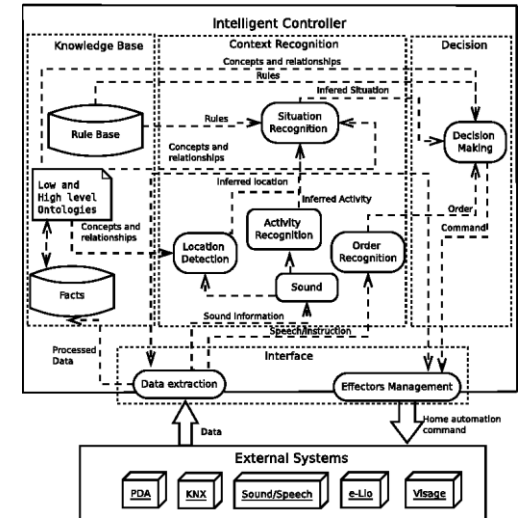


↓ data for training in NN

Symbolic vs Connectionist

Which is better?

- Future: combination; best-of-both-worlds.
 - “Neurosymbolic AI”
 - **Example:** Markov Logic Networks



Outline

- Introduction to logic
 - Arguments, validity, soundness
- Propositional logic
 - Sentences, semantics, inference
- First order logic (FOL)
 - Predicates, objects, formulas, quantifiers



Basic Logic

- Arguments, premises, conclusions 前提
 - Argument: a set of sentences (premises) + a sentence (a conclusion)
 - Validity: argument is valid iff it's necessary that if all premises are true, the conclusion is true → (The conclusion follows the premises (no matter the premise is true/false))
 - Soundness: argument is sound iff valid & premises true → (The premise is valid and true)
 - Entailment: when valid arg., premises entail conclusion → (When we have a valid argument, premises imply the conclusion)

Propositional Logic Basics

Logic Vocabulary:

- Sentences, symbols, connectives, parentheses

★ – Symbols: P, Q, R, ... (atomic sentences) → *the most fundamental sentences*

– Connectives:

$\wedge$ and

[conjunction]

$\vee$ or

[disjunction]

$\Rightarrow$ implies *(if-then)*

[implication]

$\Leftrightarrow$ is equivalent

[biconditional]

$\neg$ not

[negation]

– Literal: P or negation $\neg P$

→ *either a symbol or a negation*

Propositional Logic Basics

Examples:

- $(P \vee Q) \Rightarrow S$ $\longrightarrow$ *if- then*
 - “If it is cold or it is raining, then I need a jacket”
- $Q \Rightarrow P$
 - “If it is raining, then it is cold”
- $\neg R$
 - “It is not hot”



Propositional Logic Basics

Several rules in place

- Precedence: $\neg$, $\wedge$, $\vee$, $\Rightarrow$, $\Leftrightarrow$
- ★ • Use parentheses when needed
- Sentences: well-formed or not well-formed:
 - $P \Rightarrow Q \Rightarrow S$ **not well-formed (not associative!)**

Sentences & Semantics

- Sentences: built up from symbols with connectives

- **Interpretation:** assigning True / False → can be false/true
to symbols (a row in truth table)

- **Semantics:** interpretations for which
sentence evaluates to True → True interpretations only

- **Model:** (of a set of sentences)
interpretation for which all sentences
are True

→ (All of the assignments I can do to make all sentences true)



Another kind of model :)

Evaluating a Sentence

- Example:

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
false	false	true	false	false	true	true
false	true	true	false	true	true	false
true	false	false	false	true	false	false
true	true	false	true	true	true	true

- Note:

- ★ – If P is false, $P \Rightarrow Q$ is true regardless of Q (“5 is even implies 6 is odd” is True!)
(Just this rule, no pressure)
- Causality not needed: “5 is odd implies the Sun is a star” is True!

★ (P is false, then $P \Rightarrow Q$ is true
no matter whether Q is t/f)

Evaluating a Sentence: Truth Table

- **Ex:**

P	Q	R	$\neg P$	$Q \wedge R$	$\neg P \vee Q \wedge R$	$\neg P \vee Q \wedge R \Rightarrow Q$
0	0	0	1	0	1	0
0	0	1	1	0	1	0
0	1	0	1	0	1	1
0	1	1	1	1	1	1
1	0	0	0	0	0	1
1	0	1	0	0	0	1
1	1	0	0	0	0	1
1	1	1	0	1	1	1

- **Satisfiable**

- There exists some interpretation where the sentence is true.

→ (The sentence is satisfiable, this is different than "semantics" which refers to interpretations)

Break & Quiz

Q 1.1: Suppose P is false, Q is true, and R is true. Does this assignment satisfy

(i) $\neg(\neg p \rightarrow \neg q) \wedge r$

(ii) $(\neg p \vee \neg q) \rightarrow (p \vee \neg r)$

- A. Both
- B. Neither
- C. Just (i)
- D. Just (ii)

Break & Quiz

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Q 1.1: Suppose P is false, Q is true, and R is true. Does this assignment satisfy

(i) $\neg(\neg p \rightarrow \neg q) \wedge r$

(ii) $(\neg p \vee \neg q) \rightarrow (p \vee \neg r)$

Plug interpretation into each sentence.

- A. Both
- B. Neither
- **C. Just (i)**
- D. Just (ii)

For (i): $(\neg p \rightarrow \neg q)$ will be false so $\neg(\neg p \rightarrow \neg q)$ will be true and r is true by assignment.

For (ii): $(\neg p \vee \neg q)$ is true and $(p \vee \neg r)$ is false which makes the implication false.

Break & Quiz

Q 1.2: Let A = “Aldo is Italian” and B = “Bob is English”.
Formalize “Aldo is Italian or if Aldo isn’t Italian then Bob is English”.

- a. $A \vee (\neg A \rightarrow B)$
- b. $A \vee B$
- c. $A \vee (A \rightarrow B)$
- d. $A \rightarrow B$

Break & Quiz

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- a. $A \vee (\neg A \rightarrow B)$
- b. $A \vee B$ (equivalent!)
- c. $A \vee (A \rightarrow B)$
- d. $A \rightarrow B$

Break & Quiz

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- b. $A \vee B$ (equivalent!)
- c. $A \vee (A \rightarrow B)$
- d. $A \rightarrow B$

Answer a. is the exact translation of the English sentence into a logic sentence. You can see that answer b. is also correct by writing out the truth table for all answers and seeing that a and b have the same truth tables.

Or you can use the fact that $\neg A \rightarrow B = A \vee B$ and that $A \vee A \vee B = A \vee B$ to prove equivalence.

Knowledge Bases

A sentence
↑

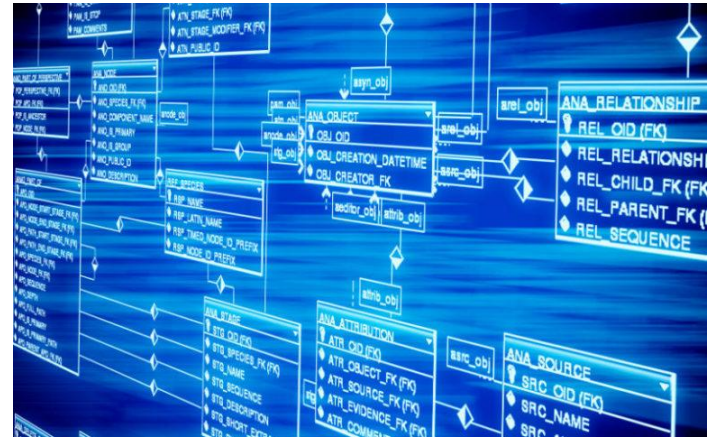
- Knowledge Base (KB): A set of sentences $\{A_1, \dots, A_n\}$
 - Like a long sentence, connect with conjunction
 - $KB: A_1 \wedge A_2 \wedge \dots \wedge A_n$

Model of a KB: interpretations where all sentences are True

★ **Goal**: inference to discover new sentences

(Remember, model means for multiple

sentences, there are some interpretations that can pt all sentences become true)



Entailment

Entailment: a sentence B logically follows from A

- Write $A \models B$ (B entails A)
- $A \models B$ iff in every interpretation where A is true, B is also true



(B can include other things, that's fine, but must cover all of A, can think of this like an inclusion relationship)

All interpretations

B is true

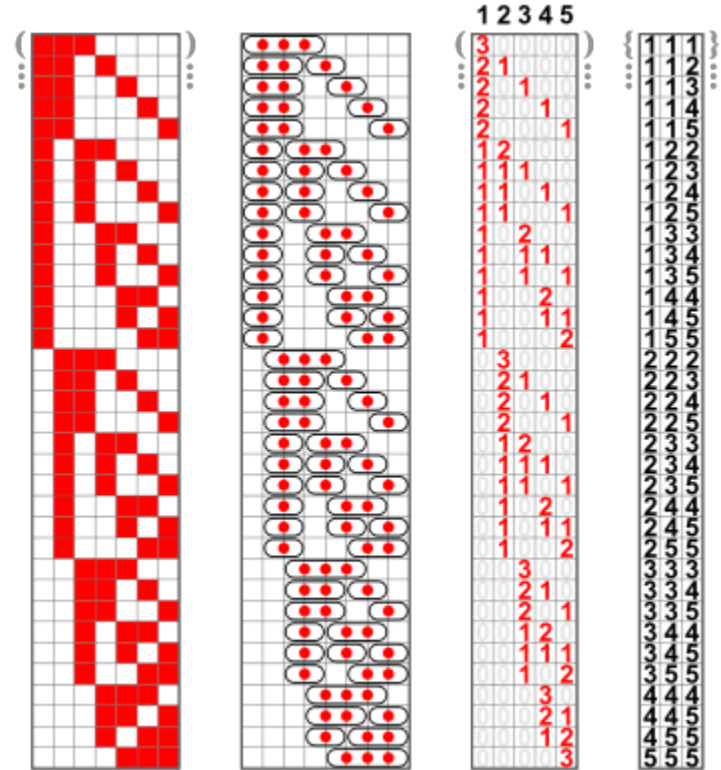
A is true

Inference

- Given a set of sentences (a KB), logical inference creates new sentences
 - Compare to prob. inference!
- **Challenges:**
 - Soundness
 - Completeness
 - Efficiency

Methods of Inference: 1. Enumeration

- Enumerate all interpretations;
look at the truth table
 - “Model checking”
- Downside: 2^n interpretations
for n symbols



Methods of Inference: 2. Using Rules

- *Modus Ponens*: $(A \Rightarrow B, A) \models B$
- And-elimination
- Other rules on the next page
 - Commutativity, associativity, de Morgan's laws, distribution for conjunction/disjunction



Logical equivalences

$$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha) \quad \text{commutativity of } \wedge$$

$$(\alpha \vee \beta) \equiv (\beta \vee \alpha) \quad \text{commutativity of } \vee$$

$$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma)) \quad \text{associativity of } \wedge$$

$$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma)) \quad \text{associativity of } \vee$$

$$\neg(\neg\alpha) \equiv \alpha \quad \text{double-negation elimination}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha) \quad \text{contraposition}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta) \quad \text{implication elimination}$$

$$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) \quad \text{biconditional elimination}$$

$$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) \quad \text{de Morgan}$$

$$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) \quad \text{de Morgan}$$

$$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) \quad \text{distributivity of } \wedge \text{ over } \vee$$

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You can use these equivalences to modify sentences.

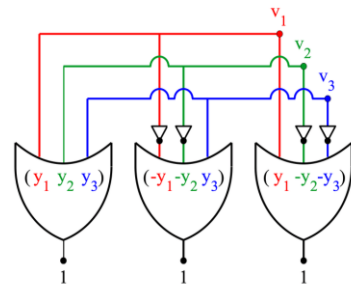
Methods of Inference: 3. Resolution

- Convert to special form and use a single rule
- **Conjunctive Normal Form (CNF)**

$$\underbrace{(\neg A \vee B \vee C)}_{\text{a clause}} \wedge (\neg B \vee A) \wedge (\neg C \vee A)$$

Conjunction of clauses; each clause disjunction of literals

- Simple rules for converting to CNF



Conjunctive Normal Form (CNF)

$$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg P_{1,2} \vee B_{1,1}) \wedge (\neg P_{2,1} \vee B_{1,1})$$

- Replace all $\Leftrightarrow$ using biconditional elimination
- Replace all $\Rightarrow$ using implication elimination
- Move all negations inward using
 - double-negation elimination
 - de Morgan's rule
- Apply distributivity of $\vee$ over $\wedge$

Convert example sentence into CNF

$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$ starting sentence

$(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$
biconditional elimination

$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg(P_{1,2} \vee P_{2,1}) \vee B_{1,1})$
implication elimination

$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge ((\neg P_{1,2} \wedge \neg P_{2,1}) \vee B_{1,1})$
move negations inward

$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg P_{1,2} \vee B_{1,1}) \wedge (\neg P_{2,1} \vee B_{1,1})$
distribute $\vee$ over $\wedge$

Resolution Steps

- Given KB and β (query)
- Add $\neg \beta$ to KB, show this leads to empty (False. Proof by contradiction)
- Everything needs to be in CNF
- Example KB:
 - $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
 - $\neg B_{1,1}$
- Example query: $\neg P_{1,2}$

Resolution Preprocessing

- Add $\neg \beta$ to KB, convert to CNF:

$$a1: (\neg B_{1,1} \vee P_{1,2} \vee P_{2,1})$$

$$a2: (\neg P_{1,2} \vee B_{1,1})$$

$$a3: (\neg P_{2,1} \vee B_{1,1})$$

$$b: \neg B_{1,1}$$

$$c: P_{1,2}$$

- Want to reach goal: *empty*

Resolution

- Take any two clauses where one contains some symbol, and the other contains its complement (negative)

PVQVR

$\neg$ QVSVT

- Merge (resolve) them, throw away the symbol and its complement

PVRVSVT

- If two clauses resolve and there's no symbol left, you have reached *empty* (False). $KB \models \beta$
- If no new clauses can be added, KB does not entail β

Resolution Example

$$a1: (\neg B_{1,1} \vee P_{1,2} \vee P_{2,1})$$

$$a2: (\neg P_{1,2} \vee B_{1,1})$$

$$a3: (\neg P_{2,1} \vee B_{1,1})$$

$$b: \neg B_{1,1}$$

$$c: P_{1,2}$$

Resolution Example

$$a1: (\neg B_{1,1} \vee P_{1,2} \vee P_{2,1})$$

$$a2: (\neg P_{1,2} \vee B_{1,1})$$

$$a3: (\neg P_{2,1} \vee B_{1,1})$$

$$b: \neg B_{1,1}$$

$$c: P_{1,2}$$

Step 1: resolve a2, c: $B_{1,1}$

Step 2: resolve above and b: *empty*

Break & Quiz

Q 2.1: Which has more rows: a truth table on n symbols, or a joint distribution table on n binary random variables?

- A. Truth table
- B. Distribution
- C. Same size
- D. It depends

Break & Quiz

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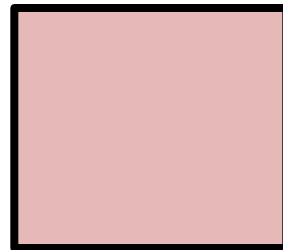
First Order Logic (FOL)

Propositional logic has some limitations

- Ex: how to say “all squares have four sides”
- No context, hard to generalize; express facts

FOL is a more expressive logic; works over

- Facts, Objects, Relations, Functions



First Order Logic Syntax

- **Term:** an object in the world
 - **Constant:** Alice, 2, Madison, Green, ...
 - **Variables:** x, y, a, b, c, ...
 - **Function**(term₁, ..., term_n)
 - Sqrt(9), Distance(Madison, Chicago)
 - Maps one or more objects to another object
 - Can refer to an unnamed object: LeftLeg(John)
 - Represents a user defined functional relation
- A **ground term** is a term without variables.
 - Constants or functions of constants

FOL Syntax

- **Atom**: smallest T/F expression
 - **Predicate**(term₁, ..., term_n)
 - Teacher(Jerry, you), Bigger(sqrt(2), x)
 - Convention: read “Jerry (is)Teacher(of) you”
 - Maps one or more objects to a truth value
 - Represents a user defined relation
 - **term₁ = term₂**
 - Radius(Earth)=6400km, 1=2
 - Represents the equality relation when two terms refer to the same object

FOL Syntax

- **Sentence:** T/F expression
 - Atom
 - Complex sentence using connectives: $\wedge \vee \neg \Rightarrow \Leftrightarrow$
 - $\text{Less}(x,22) \wedge \text{Less}(y,33)$
 - Complex sentence using quantifiers $\forall, \exists$
- Sentences are evaluated under an interpretation
 - Which objects are referred to by constant symbols
 - Which objects are referred to by function symbols
 - What subsets defines the predicates

FOL Quantifiers

- Universal quantifier: $\forall$
- Sentence is true **for all** values of x in the domain of variable x .
- Main connective typically is $\Rightarrow$
 - Forms if-then rules
 - “all humans are mammals”
$$\forall x \text{ human}(x) \Rightarrow \text{mammal}(x)$$
 - Means if x is a human, then x is a mammal

FOL Quantifiers

- Existential quantifier: $\exists$
- Sentence is true **for some** value of x in the domain of variable x .
- Main connective typically is $\wedge$
 - “some humans are male”
 $\exists x \text{ human}(x) \wedge \text{male}(x)$
 - Means there is an x who is a human and is a male

Break & Quiz

Q 2.1: How many entries does a truth table have for a FOL sentence with k variables where each variable can take on n values?

- A. Truth tables are not applicable to FOL.
- B. 2^k
- C. n^k
- D. It depends

Break & Quiz

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Must have one entry for every possible assignment of values to variables. That number is (C).